SAI VENKATESH PALATLA

Dallas, Texas | +1-(XXX) XXX-XXXX | saivpalatla@gmail.com (preferred, fastest response)
LinkedIn: saivenkateshp | GitHub: saivenkateshp

SUMMARY

AI/ML Engineer with 3+ years of experience building Generative AI applications and scalable data platforms on Google Cloud (BigQuery, GKE, Vertex AI). Experienced in designing RAG pipelines, orchestrating agentic AI workflows, and translating business problems into production solutions for stakeholders.

Core Competencies: Cloud Platforms, GenAI Orchestration, Distributed Data Pipelines, Microservices Architecture, API Development.

WORK EXPERIENCE

Verizon Communications Inc. | Dallas, Texas, USA
Software Development Engineer / AI Engineer | February 2024 - April 2026

- Built and deployed a real-time Retrieval-Augmented Generation (RAG) pipeline ingesting network documents, runbooks, incident tickets, and logs (chunking, embeddings, Pinecone vector search), served through FastAPI and powered by the Gemini API, reducing incident troubleshooting time.
- Developed agentic AI workflows using LangChain and Google ADK, applying prompt engineering and output guardrails for text classification, summarization, semantic search, and document understanding for internal users.
- Leveraged dynamic prompt engineering and AI-assisted development tools (including GitHub Copilot, Claude) to accelerate the design, testing, and deployment of backend applications.
- Designed and deployed executive-level data visualizations and interactive dashboards using Tableau and Power BI to translate complex data models into actionable business insights.
- Engineered distributed data processing pipelines using Apache Spark (PySpark) on GCP Dataproc to extract, transform, and load high-volume network telemetry data from Cloud Storage (GCS), significantly reducing batch processing times.
- Architected scalable dimensional data models (star schemas) in BigQuery, developing complex workflows to populate fact tables and slowly changing dimensions (SCDs) for optimized historical reporting and downstream analytics.
- Developed high-performance backend microservices and REST/GraphQL APIs using FastAPI and Spring Boot to securely expose curated metrics to downstream applications.
- Investigated and resolved complex ingestion and transformation failures across distributed pipelines and services, meeting strict SLA targets and ensuring data security.
- Implemented comprehensive CI/CD pipelines, establishing robust monitoring systems using Grafana and Prometheus to ensure high observability and reliability for production APIs.
- Enforced least-privilege access with RBAC via GCP IAM roles and service accounts across cloud resources and data assets, supporting data privacy, governance, and compliance standards.

Personalis Inc. | Menlo Park, California, USA
Data Engineer Intern | June 2022 - January 2023

- Engineered automated ETL workflows using Python, Pandas, and SQL to extract, validate, and load complex finance and sales data from SAP and other disparate source systems.
- Built incremental ingestion pipelines with Airflow and BigQuery SQL MERGE logic to upsert daily records into the master enterprise warehouse.

- Improved pipeline efficiency by 50% and data quality to 95% across thousands of financial transactions through workflow orchestration and validation controls.
- Optimized database performance by implementing a retention strategy, archiving stale data to GCS buckets to support a modern cloud architecture.
- Developed containerized Flask microservices and GraphQL APIs on GKE to support reliable data access and downstream integrations, enhancing deployment reliability by 90%.
- Established a centralized BigQuery Data Catalog, streamlining metadata management to improve data discoverability and support self-service analytics for over $50M+ in sales.

IBM India Software Labs Pvt Ltd | Bengaluru, Karnataka, India
Software Development Engineer Intern | January 2019 - July 2019

- Developed an innovative testing framework in Java, JavaScript, and Loopback to rigorously validate each connector and ensure seamless, accurate data flow between integrated applications.
- Achieved a 90% decrease in integration errors and significantly boosted testing speed and reliability through automated connector validation.
- Pioneered the integration of Swagger for API standards, resulting in a 60% decrease in design time while enhancing the accuracy and scalability of API documentation.
- Formulated robust test cases for microservices within an automated Jenkins CI/CD pipeline, resulting in a 70% decrease in SDLC duration and a streamlined deployment process.
- Spearheaded the resolution of complex production tickets and reduced the legacy ticket backlog by 40% through active collaboration with cross-functional teams to identify root causes and deploy effective fixes.

TECHNICAL SKILLS

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Agentic AI Orchestration, Google ADK, LangChain, LangGraph, Gemini API, OpenAI API, Vector Databases (Pinecone), Prompt Engineering, Natural Language Processing (NLP), AI-Assisted Development (GitHub Copilot, Claude, Cursor, ChatGPT)

Programming Languages: Python, SQL, Java, JavaScript, C, C++, R, HTML/CSS

Data Engineering & Streaming: Snowflake, Apache Airflow, Apache Spark, Apache Kafka, ETL/ELT, Data Modeling

Cloud Technologies: GCP (BigQuery, GCS, GKE, Vertex AI, Cloud Composer, Cloud Functions, Dataproc), AWS (EC2, S3, CloudWatch)

Data Platforms & Tooling: Databricks, ADF (Azure Data Factory), Lakehouse Architectures, Orchestration

APIs, Frameworks & Services: FastAPI, Spring Boot, Flask, Django, Node.js, REST APIs, GraphQL, Microservices

Databases & Search: BigQuery, MySQL, MongoDB, Elasticsearch, Apache Solr

DevOps & BI Tools: Docker, Kubernetes, Jenkins, Grafana, Prometheus, Looker, Tableau, Power BI, Kibana, Git

EDUCATION

University of Texas at Dallas | August 2021 - May 2023
Master of Science in Computer Science

Amrita Vishwa Vidyapeetham | July 2015 - July 2019
Bachelor of Engineering in Computer Science